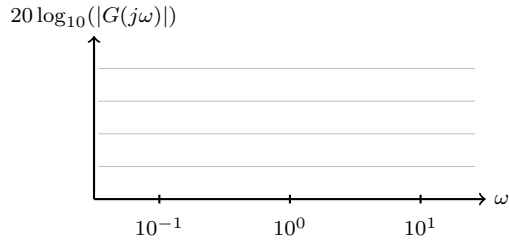


1. Open-loop plant

$$G(s) = \frac{5}{s+2}$$

Sketch the Bode magnitude plot for the transfer function above.



Questions

- What is the DC gain? _____
- What is the open-loop bandwidth? _____

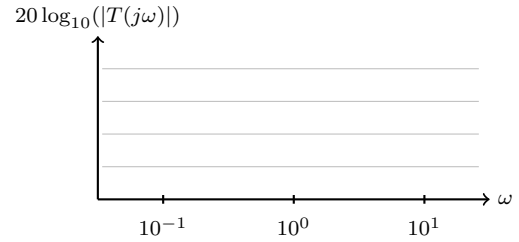
2. P-only controller

Choose k_p : $k_p =$ _____

$$T(s) = \frac{G(s)C(s)}{1 + G(s)C(s)}$$

$T(s) =$ _____

Sketch the Bode magnitude plot for the closed-loop transfer function.



Questions

- What is the DC gain? _____
- What is the closed-loop bandwidth? _____

3. PI controller

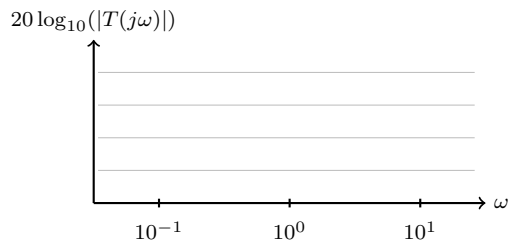
Choose k_p and k_i :

$$C(s) = k_p + \frac{k_i}{s}, \quad k_p = \text{_____,} \quad k_i = \text{_____}$$

$$T(s) = \frac{G(s)C(s)}{1 + G(s)C(s)}$$

$T(s) =$ _____

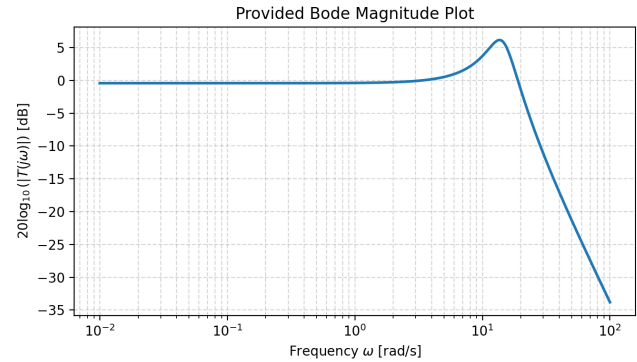
Sketch the Bode magnitude plot for the closed-loop transfer function.



Questions

- What is the DC gain? _____
- What is the closed-loop bandwidth? _____

4. Provided Bode plot



Questions

- What is the DC gain? _____
- Effect of controller: _____